

VC DIMENSION UPPER-BOUNDS RADEMACHER COMPLEXITY

A CLASSICAL ROUTE FROM INFINITE HYPOTHESIS CLASSES TO UNIFORM GENERALISATION

Jay Parekh, Yeshwanth Patil, Andrew Robertson

COMP34312, The University of Manchester

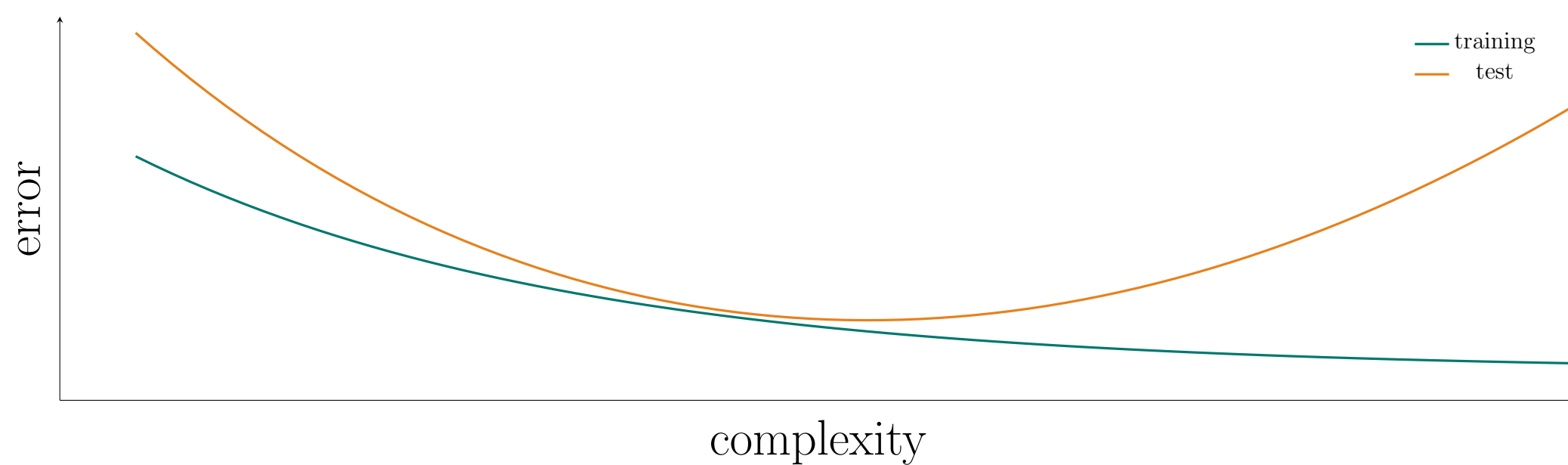
1. Problem: Training Error is Not Enough

Goal. We minimise empirical risk, but care about population risk:

$$R(f) = \mathbb{E}_{(x,y) \sim D}[\ell(y, f(x))], \quad \widehat{R}(f) = \frac{1}{m} \sum_{i=1}^m \ell(y_i, f(x_i)).$$

Question. When is $\widehat{R}(f)$ a reliable proxy for $R(f)$ uniformly over $f \in \mathcal{F}$?

$$\sup_{f \in \mathcal{F}} (\widehat{R}(f) - R(f))$$



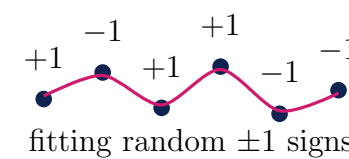
Message. A class that is too flexible can fit training data without generalising.

2. Rademacher Complexity: Can the Class Fit Noise?

Let $\epsilon_i \in \{-1, +1\}$ be independent random signs. For real-valued $h \in \mathcal{H}$,

$$\mathfrak{R}_m(\mathcal{H}) = \mathbb{E}_{S, \epsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \epsilon_i h(z_i) \right].$$

Intuition. The supremum is large when some h aligns with pure random noise.



3. Master Theorem: Complexity Controls the Gap

Symmetrisation gives

$$\mathbb{E}_S \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{m} \sum_{i=1}^m h(z_i) - \mathbb{E}[h(z)] \right) \right] \leq 2\mathfrak{R}_m(\mathcal{H}).$$

Proof idea. Introduce an independent “ghost sample”, replace the population expectation by its empirical average, then use random signs to symmetrise the two samples.

Bound $\mathfrak{R}_m(\mathcal{H}) \Rightarrow$ bound generalisation.

4. From Functions to Label Vectors

For $\mathcal{F} \subseteq \{\pm 1\}^{\mathcal{X}}$ and fixed points $x_1, \dots, x_m$,

$$\mathcal{F}|_{x_1^m} = \{(f(x_1), \dots, f(x_m)) : f \in \mathcal{F}\} \subseteq \{\pm 1\}^m.$$

Key idea. $|\mathcal{F}|_{x_1^m}|$ counts how many different labelings the class can produce on this dataset.

Multiple functions can produce the same label vector on a fixed dataset.

5. Growth Function: Worst-Case Number of Labelings

The growth function is

$$\Pi_{\mathcal{F}}(m) = \max_{x_1, \dots, x_m \in \mathcal{X}} |\mathcal{F}|_{x_1^m}|$$

It asks: what is the largest number of distinct binary labelings the class can realise on any m points?

$$\Pi_{\mathcal{F}}(m) \leq 2^m.$$

For $m = 2$, there are at most $2^2 = 4$ possible labelings.

6. Massart’s Lemma: Finite Vectors Bound Rademacher Complexity

For finite $A \subset \mathbb{R}^m$ with $r = \sup_{a \in A} \|a\|_2$,

$$\mathbb{E}_{\epsilon} \left[\sup_{a \in A} \frac{1}{m} \sum_{i=1}^m \epsilon_i a_i \right] \leq \frac{r \sqrt{2 \ln |A|}}{m}.$$

Apply to $A = \mathcal{F}|_{x_1^m}$. Since each binary label vector has $r = \sqrt{m}$,

$$\mathfrak{R}_m(\mathcal{F}) \leq \sqrt{\frac{2 \ln \Pi_{\mathcal{F}}(m)}{m}}.$$

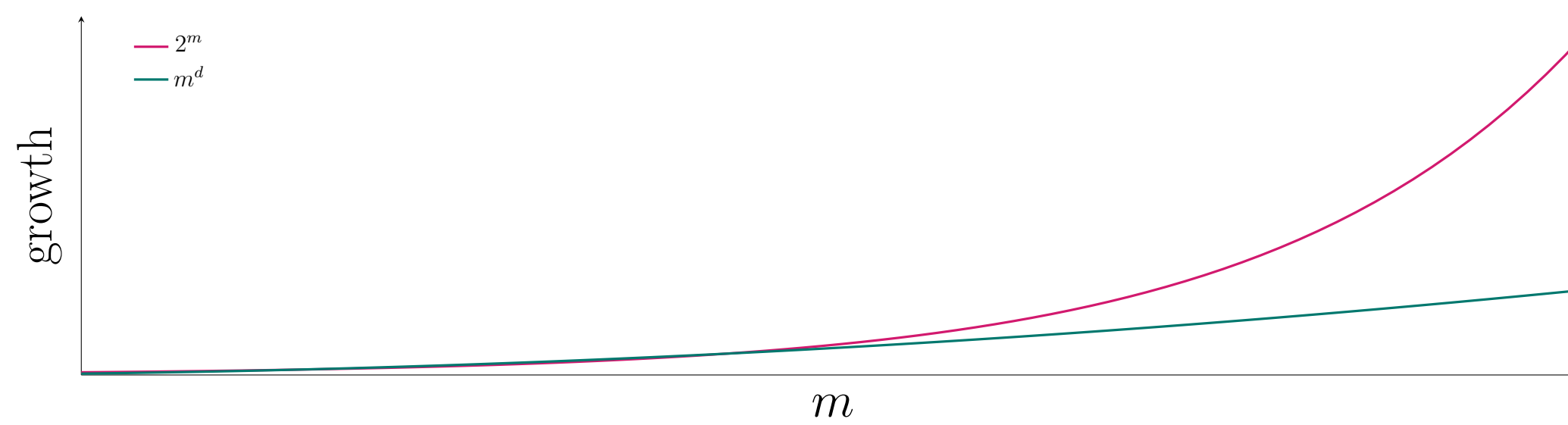
Rademacher complexity is controlled by the growth function

7. Why the Trivial Bound Fails

Using only $\Pi_{\mathcal{F}}(m) \leq 2^m$ gives

$$\mathfrak{R}_m(\mathcal{F}) \leq \sqrt{\frac{2 \ln(2^m)}{m}} = \sqrt{2 \ln 2}.$$

This constant does not vanish as $m \rightarrow \infty$, so it cannot prove that more data improves generalisation.



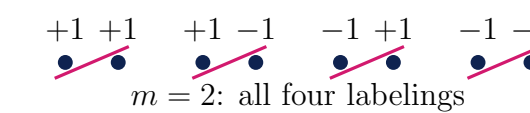
8. VC Dimension: Where Full Flexibility Stops

A set $\{x_1, \dots, x_m\}$ is **shattered** by $\mathcal{F}$ if every binary labeling can be realised:

$$\mathcal{F}|_{x_1^m} = \{\pm 1\}^m.$$

The **VC dimension** is

$$\text{VCdim}(\mathcal{F}) = \max\{m : \Pi_{\mathcal{F}}(m) = 2^m\}.$$



Class $\mathcal{F}$	$\text{VCdim}(\mathcal{F})$
Intervals on $\mathbb{R}$	2
Rectangles in $\mathbb{R}^2$	4
Halfspaces in $\mathbb{R}^d$	$d + 1$
Convex polygons in $\mathbb{R}^2$	∞

9. Sauer’s Lemma: VC Dimension Controls Growth

If $\text{VCdim}(\mathcal{F}) = d$, then

$$\Pi_{\mathcal{F}}(m) \leq \sum_{i=0}^d \binom{m}{i}.$$

For $m \geq d$,

$$\Pi_{\mathcal{F}}(m) \leq \left(\frac{em}{d} \right)^d.$$

Meaning. After the class stops shattering beyond d points, growth is polynomial instead of exponential:

$$2^m \text{ becomes roughly } m^d.$$

10. Final Chain: VC Dimension Upper-Bounds Rademacher Complexity

For $m \geq d$,

$$\mathfrak{R}_m(\mathcal{F}) \leq \sqrt{\frac{2 \ln \Pi_{\mathcal{F}}(m)}{m}} \leq \sqrt{\frac{2d \ln(em/d)}{m}}.$$

Therefore,

$$\mathbb{E}_S \left[\sup_{f \in \mathcal{F}} (\widehat{R}(f) - R(f)) \right] \lesssim \sqrt{\frac{d \ln(em/d)}{m}}.$$

Interpretation.

- Larger VC dimension d means a richer class.
- Larger sample size m improves generalisation.
- Finite VC dimension gives uniform convergence.

11. Takeaway and References

A class generalises when it cannot realise too many labelings relative to the amount of data.



References: COMP34312 course notes; Kakade & Tewari Lectures 10–11; Vapnik–Chervonenkis (1971); Sauer (1972).